From Classroom/Integrated School Programs 7 in Top 20, 23 in Top 100, 54 in Top 300, 106 in Top 500 All India Ranks & 2314 Students from Classroom /Integrated School Programs & 3723 Students from All Programs have been Awarded a Rank in JEE (Advanced), 2013

JEE(Main)-2014

ANSWERS, HINTS & SOLUTIONS

PART TEST–IV

(Main)

Q. No.	PHYSICS	CHEMISTRY	MATHEMATICS
1.	B	C	C
2.	B	B	C
3.	A	A	D
4.	A	A	A
5.	B	A	A
6.	B	D	C
7.	B	C	A
8.	D	D	B
9.	C	D	A
10.	C	A	D
11.	C	D	B
12.	C	A	A
13.	C	B	D
14.	A	D	B
15.	D	C	D
16.	B	A	A
17.	A	A	C
18.	A	B	A
19.	B	B	C
20.	D	C	B
21.	B	C	B
22.	C	B	B
23.	B	D	C
24.	B	C	D
25.	D	B	C
26.	A	B	D
27.	A	C	C
28.	C	D	D
29.	A	B	C
30.	B	D	D

Physics

PART – I

SECTION – A

1. use $\frac{1}{v}+\frac{1}{u}=\frac{2}{R}$. Image after 1st reflection will act as object for 2nd reflection.
2. In inductor current lags voltage by $\pi/2$
3. $$n(4I_0)=\frac{\int_{y=-\frac{\lambda D}{4d}}^{y=\frac{\lambda D}{4d}} 4I_0\cos^2\frac{\pi dy}{\lambda D}dy}{\int_{-\frac{\lambda D}{4d}}^{\frac{\lambda d}{4d}} dy}$$
4. $$E=\frac{hc}{\lambda}=\frac{GM^2}{L}$$
   $\Rightarrow$ dimensions of $\frac{hc}{G}=\frac{LM^2}{L}=M^2$
5. $\langle P\rangle = V_{RMS}I_{RMS}\cos\phi$
11. Use Lenz's Rule
12. Speed of image = speed of object in all the mirrors
13. Use $\mu=\frac{\sin\left(\frac{A+\delta_m}{2}\right)}{\sin\left(\frac{A}{2}\right)}$
15. $\varepsilon=-\frac{d\phi}{dt}$ and $P=\frac{\varepsilon^2}{R}$
16. For CBF $\Delta P = 0 = t_A(\mu_A-1)-t_B(\mu_B-1)+\frac{dy}{D}$
18. $V_A - iR + 15 + (5\times10^{-3})\times10^3 = V_B$

Chemistry

PART – II

SECTION – A

2. $\frac{d_{bcc}}{d_{ccp}} = \frac{\text{packing efficiency of bcc}}{\text{packing efficiency of ccp}} = \frac{67.92}{74.02} = 0.918$

3. Except Ti(+IV) in $TiCl_4$ all other metal ions Fe^{3+} in $FeCl_3$, Cu^{2+} in $CuCl_2$ and Mn^{2+} in $MnCl_2$ contains unpaired electrons and are coloured.

4. Fe^{2+} = 0.07 mole in 0.1 mol of $Fe_{0.9}O$
So to balance charge, some of the place has been left vacant as Fe^{2+} is replaced by Sr^{4+}.
$0.07 \times 2 = x \times 4$
$x = \frac{0.07 \times 2}{4} = 0.035$
Vacant place = $0.07 - 0.035 = 0.035\ N_A$

5. In diamond, all the carbon atoms are tetrahedrally bonded to one another so octahedral voids cannot be occupied. Also, the lattice points and the holes are occupied by carbon atoms. As a result the packing efficiency is less than 74%.

7. $4FeCl_3 + 3K_4[Fe(CN)_6] \longrightarrow \underset{\text{Prussian blue}}{Fe_4[Fe(CN)_6]_3} + 12KCl$

8. $SnCl_2 + HgCl_2 \longrightarrow SnCl_4 + \underset{(\text{white})}{Hg_2Cl_2}\downarrow$

$Hg_2Cl_2 + SnCl_2 \longrightarrow \underset{(\text{grey})}{Hg}\downarrow + SnCl_4$

9. $\frac{Y_A}{Y_B} = \frac{P_A / P_T}{P_B / P_T} = \frac{\chi_A . P_A^0}{\chi_B . P_B^0}$

$\Rightarrow \frac{Y_A}{\chi_A} = \frac{Y_B}{\chi_B}\left(\frac{P_A^0}{P_B^0}\right)$

Because $\frac{P_A^0}{P_B^0} > 1$

Hence $\frac{Y_A}{Y_B} > \frac{\chi_A}{\chi_B}$

10. $PBr_5(s) \longrightarrow [PBr_4]^+(s) + Br^-(s); \quad [i = 2]$

$2PCl_5(s) \longrightarrow [PCl_6]^- + [PCl_4]^+; \quad [i = 1]$

ratio $= \frac{2}{1} = 2$

11. $\pi = \frac{CRT}{M}$

$\frac{RT}{M} = 4.65 \times 10^{-3}$

$8.314 \times \frac{293}{M} = 4.65 \times 10^{-3}$

$M = 5.24 \times 10^5$

12. $3.6 \times 10^{-7} = \frac{2s}{55.55}$

(Let solubility is 's')

Solubility = 10^{-5}

13. $P_S = \frac{80P^o}{100}$, W_B = ?, M_B = 40, W_A = 114 g, M_A = 114 g

$\Delta P = 0.2\ P^0$

Hence, $0.2 = \frac{W_B / 40}{(W_B / 40) + 1}$

Hence W_B = 10 gm

15. $E^o = \frac{0.059}{n} \log_{10} K$

$= \frac{0.059}{2} \log_{10} 10^8$

= 0.236

$E^o_{Cell} = E^o_{Reduced\ species} - E^o_{Oxidised\ species}$

$0.236 = E^o_{Fe^{3+}/Fe^{2+}} - 0.54$

$E^o_{Fe^{3+}/Fe^{2+}} = 0.776V$

16. $Pt \mid H_2 \mid H^+ \parallel H^+ \mid H_2 \mid Pt$

1 atm x 10^{-4} 1 atm

$H_2(g) \longrightarrow 2H^+ + 2e^-$; at anode

x

$2H^+ + 2e^- \longrightarrow H_2(g)$; at cathode

10^{-4}M

$E_{cell} = \frac{0.0591}{2} \log \frac{(10^{-4})^2}{x^2}$

$x = 10^{-6}$ M = $[H^+]$

19. Copper deposited = 0.01 mole; Cu^{2+} already present = 0.05 mole.
Total solute remaining = 0.04 + 0.05 = 0.09 mole in 250 ml (250 gm) H_2O.

$\Delta T_f = K_f.m = 1.86 \times \frac{0.09}{0.25} = 0.67$

$T_f = -0.67$

21. $(NH_4)_2Cr_2O_7 \xrightarrow{\Delta} N_2 + Cr_2O_3 + 4H_2O$

$NH_4NO_2 \xrightarrow{\Delta} N_2 + 2H_2O$

22. Zinc blende is ZnS and not $ZnCl_2$.

23. Aluminium will react with carbon to form aluminium carbide.

26. $Bi^{3+} + KI \longrightarrow \underset{\text{(brownish black)}}{BiI_3 \downarrow}$

$BiI_3 \xrightarrow{KI} \underset{\text{(orange yellow solution)}}{[BiI_4]^-}$

27. Nitroprosside ion is $[Fe(CN)_5NO]^{2-}$. If the central atom is present here in Fe^{2+} form, its effective atomic number will be $26 - 2 + 6 \times 2 = 36$.
It does not have unpaired electron, so this anionic complex is diamagnetic. If the nitroprusside ion is having Fe^{3+} and NO, the electronic distribution will be such that it will have one unpaired electron and the complex will be paramagnetic. Thus magnetic moment measurement established that in nitroprusside ion, the Fe and NO exist Fe^{II} and NO^+ rather the Fe^{III} and NO.

29. $ICl_3 \xrightarrow{\text{hydrolysis}} HIO_2 + HCl$

$IF_5 \xrightarrow{\text{hydrolysis}} HIO_3 + HF$

$IF_7 \xrightarrow{\text{hydrolysis}} HIO_4.2H_2O + HF$

$ICl \xrightarrow{\text{hydrolysis}} HOI + HCl$

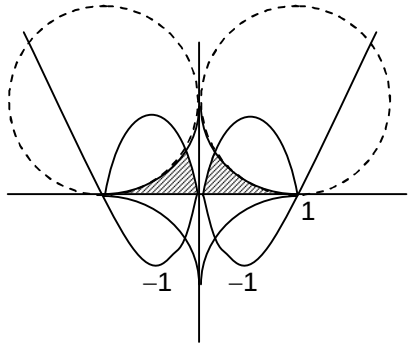
30. P_4O_{10} is anhydride of H_3PO_4.
All P atom are sp^3 hybridized.
There are 6 (P — O — P) linkage in one molecule.

Mathematics PART – III

SECTION – A

1. There should not be a gap of 5 or more than 5 between any two numbers from the set {3, 5, 6, 9}. This gap can occur only when these numbers can occupy first four and last four position.
Hence number of favourable ways = $9! - 2(4! \times 5!)$
Total number of cases are 9!
Hence probability = $1-\frac{2\times 4!\times 5!}{9!}=\frac{62}{63}$

2. Required area is
$$2\left[\int_0^{\alpha} ex\ln x+\int_1^{\alpha}\left(1-\sqrt{1-(x-1)^2}\right)dx\right]$$



3. Let $\bar{d} = x\left(\bar{b}\times\bar{c}\right)+y\left(\bar{c}\times\bar{a}\right)+z\left(\bar{a}\times\bar{b}\right)$
Where $x=\frac{\vec{a}\cdot\vec{d}}{\left[\bar{a}\bar{b}\bar{c}\right]}$, $y=\frac{\bar{b}\cdot\bar{d}}{\left[\bar{a}\bar{b}\bar{c}\right]}$, $z=\frac{\bar{c}\cdot\bar{d}}{\left[\bar{a}\bar{b}\bar{c}\right]}$
$\Rightarrow \left[\bar{a}\bar{b}\bar{c}\right]\bar{d}=\left(\bar{a}\cdot\bar{d}\right)\left(\bar{b}\times\bar{c}\right)+\left(\bar{b}\cdot\bar{d}\right)\left(\bar{c}\times\bar{d}\right)+\left(\bar{c}\cdot\bar{d}\right)\left(\bar{a}\times\bar{b}\right) \Rightarrow |\bar{d}| = 1.$

4. AB = A + B
$\Rightarrow$ B = AB – A
= A (B – I)
$\Rightarrow$ det (B) $\equiv$ det A · det (B – I) = 0
$\Rightarrow$ det (B) = 0.

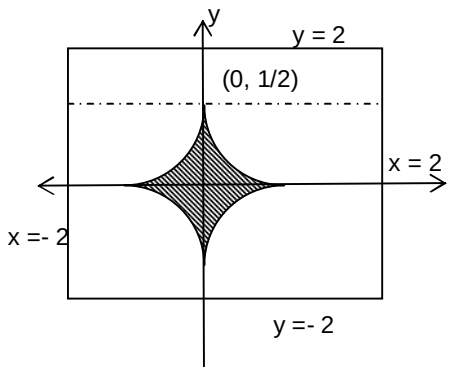
5. Given equation can be written as
$2x^2y\,(x^2dy + 2xy\,dx) + y^2\,dy - x^2\,dx = 0$
or $2x^2yd\,(x^2y) + y^2\,dy - x^2\,dx = 0$
Integrating, we get
$3\,(x^2y)^2 + y^3 - x^3 = c$

6. Area of the triangle inscribed in the ellipse is maximum if difference of the eccentric angles of the point A, B, C is $\frac{2\pi}{3}$.
So maximum area of the inscribed triangle is $\frac{3\sqrt{3}}{4}\cdot\frac{1}{6}\cdot 8\sqrt{3}=3$ sq units.

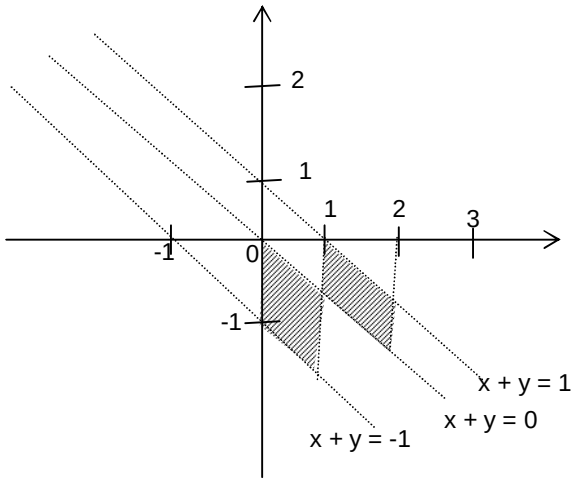
7. PM = $\sqrt{\left(1-\frac{1}{2}\right)^2+(0-1)^2\left(2+\frac{3}{2}\right)^2}=\frac{3\sqrt{6}}{2}$

8. Take $e^{-i\pi/3}$ common from R_2 and multiply in C_2.

9. **A**



10. If $x \in (0, 1)$
Then $-1 \le x + y < 0$
and if $x \in (1, 2)$
$0 \le x + y < 1$
Required area = $4\left(\frac{1}{2}\cdot 1\cdot \sqrt{2}\sin\frac{\pi}{4}\right)$ = 2 sq units.



11. Writing $y = vx$ in the given equations, we get

$$x\frac{dv}{dx} = \phi\left(\frac{1}{v}\right)$$

and $\log c + \log x = \frac{1}{v}$

$$\Rightarrow -v^2 = x\frac{dv}{dx}$$

$$\Rightarrow \phi\left(\frac{1}{v}\right) = -v^2$$

$$\Rightarrow \phi\left(\frac{x}{y}\right) = -\frac{y^2}{x^2}.$$

12. D.C of the line PQ are $\left(\frac{2}{\sqrt{17}}, \frac{-3}{\sqrt{17}}, \frac{2}{\sqrt{17}}\right)$ and those of the normal to the given plane are $\left(\frac{2}{3}, \frac{-1}{3}, \frac{-2}{3}\right)$.

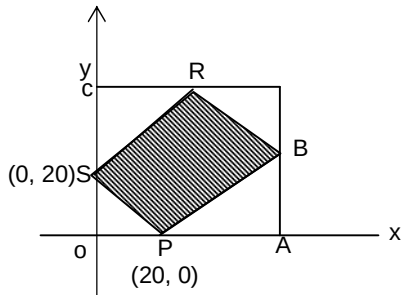
If θ is the angle between these two lines, then $\cos\theta = \frac{1}{\sqrt{17}}$

$\Rightarrow$ the angle ϕ between PQ and the plane is given by $\cos\phi = \frac{4}{\sqrt{17}}$.

Length PQ = $\sqrt{17}$.

Hence the length of the projection of PQ on the plane = $\sqrt{17}\cdot\frac{4}{\sqrt{17}} = 4$.

13. Let Ram arrives x minutes and Shyam y minutes after 11 noon. So they will be able to meet if $|x - y| \le 20$. Thus
Favourable cases = Area OPQBRSO
Total ways = Area OABC
$\frac{2000}{3600} = \frac{5}{9}$.



14. Let $x^2 - 13 = t$, then
$t^3 - 67t + 126 = 0$
$\Rightarrow t = -9, 2, 7 \Rightarrow x = \pm 2, \pm\sqrt{20}, \pm\sqrt{15}$

15. $\vec{a}\times\vec{b} - 2\vec{a}\times\vec{c} = \vec{0} = \vec{a}\times\left(\vec{b} - 2\vec{c}\right) = \vec{0} \Rightarrow \vec{a}$ and $\vec{b} - 2\vec{c}$ are collinear

$\Rightarrow b - 2\vec{c} = \lambda\vec{a} \Rightarrow |\vec{b} - 2\vec{c}| = \lambda^2 |\vec{a}|^2 \Rightarrow |\vec{b}|^2 + 4|\vec{c}|^2 - 4\vec{b}\cdot\vec{c} = \lambda^2 |\vec{a}|^2$

$\Rightarrow 16 + 4 - 4\,(4)\,(1)\,.\,\frac{1}{4} = \lambda^2\,(1)$

$\Rightarrow \lambda = \pm 4$.

16. $A^2 = \begin{bmatrix} ab & b^2 \\ -a^2 & -ab \end{bmatrix}\begin{bmatrix} ab & b^2 \\ -a^2 & -ab \end{bmatrix} = \begin{bmatrix} a^2b^2 - a^2b^2 & ab^3 - ab^3 \\ -a^3b + a^3b & -a^2b^2 + a^2b^2 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

$\therefore$ Minimum value of n = 2

17. $-d\left(\frac{x}{y}\right) = xy(1+\log x)dx$

$\int -\frac{x}{y}d\left(\frac{x}{y}\right) = \int x^2(1+\log x)dx$ gives solution.

18. y_1 is solution of $\frac{dy}{dx} + Py = Q$ is $\frac{dy_1}{dx} + Py_1 = Q$ …(1)
again $y_2 = y_1 z$ is solution of (1)
then $\frac{d}{dx}(y,\ z) + P(y,\ z) = Q$ …(2)
(2) – (1) $\times$ z
$\Rightarrow z = 1 + ae^{-\int\frac{Q}{y_1}dx}$.

20. $P = \frac{{}^{52}C_1\,{}^{5}C_2\,2!}{\left({}^{52}C_2\right)^2} = \frac{50}{663}$

21. Required area A = $\int_{\pi}^{2\pi}((\sin x - x) + 2\pi)dx = \frac{\pi^2}{2} - 2$ sq. units.

22. Area = $\pi ab = \pi \sin\alpha \operatorname{cosec}\alpha = \pi$.

23. $|\bar{r} \times \bar{a}|^2 + |\bar{r} \times \bar{b}|^2 + |\bar{r} \times \bar{c}|^2 = \sin^2\alpha + \sin^2\beta + \sin^2\gamma = 2$

24. n(S) = 81, E = sum of number is 11
n (E) = 9 $\Rightarrow$ P(E) = 9/81 = 1/9.

25. Pre-multiply the given equation by $\begin{bmatrix} 5 & 0 \\ -a & 5 \end{bmatrix}$, we get

$$\begin{bmatrix} 5 & 0 \\ -a & 5 \end{bmatrix}\begin{bmatrix} \frac{1}{25} & 0 \\ x & \frac{1}{25} \end{bmatrix} = \begin{bmatrix} 5 & 0 \\ -a & 5 \end{bmatrix}^{-1}$$

$$\Rightarrow \begin{bmatrix} 5 & 0 \\ -a & 5 \end{bmatrix}\begin{bmatrix} 5 & 0 \\ -a & 5 \end{bmatrix}\begin{bmatrix} \frac{1}{25} & 0 \\ x & \frac{1}{25} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 1 & 0 \\ -\frac{2a}{5} + 25x & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$\Rightarrow 25x - \frac{2a}{5} = 0.$

26. $\because$ A is a unit matrix
so $A^2 = A$.

27. $2x^3y\,dy + (1 - y^2)(x^2y^2 + y^2 - 1)dx = 0$

$$\frac{2y}{(1-y^2)^2}\frac{dy}{dx} + \frac{y^2}{1-y^2}\frac{1}{x} = \frac{1}{x^3}$$

Put $\frac{y^2}{1-y^2} = u \Rightarrow \frac{2y}{(1-y^2)^2}\frac{dy}{dx} = \frac{du}{dx}$

$$\Rightarrow \frac{du}{dx} + \frac{u}{x} = \frac{1}{x^3}$$

$u.x = \int \frac{1}{x^2} x\,dx + c \Rightarrow x^2y^2 = (cx - 1)(1 - y^2).$

28. $\bar{a} \times \bar{b} = \hat{i} - \hat{j} + \hat{k}$

$\therefore \bar{c} = \pm\frac{\hat{i} - \hat{j} + \hat{k}}{\sqrt{3}}.$

30. Equation of circle will be $(x + r)^2 + y^2 = r^2$.